



Fractionation of sugarcane bagasse using hydrothermal and advanced oxidative pretreatments for bioethanol and biogas production in lignocellulose biorefineries

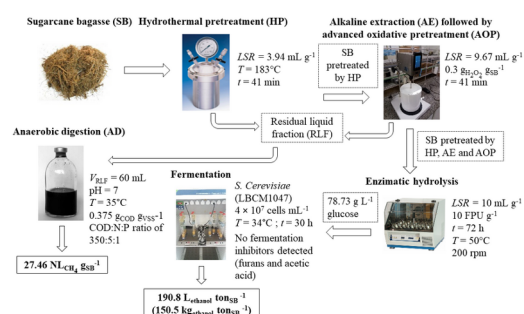
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GRAPHICAL ABSTRACT



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ABSTRACT

The fractionation of sugarcane bagasse (SB) by hydrothermal pretreatment (HP, autohydrolysis) followed by alkaline extraction (AE) and advanced oxidative pretreatment (AOP) for production of second-generation ethanol and biogas was investigated. The AOP of SB was optimized using a Doehlert design, varying the applied H₂O₂ load, liquid-to-solid ratio (LSR), and time. The responses evaluated were yield (Y), residual cellulose (RC), delignification (DE), and enzymatic conversion (EC). The AE of SB pretreated by HP led to 61.8% DE (using 0.2 mol L⁻¹ NaOH). This high lignin removal enabled substantial savings of H₂O₂ in the AOP. The optimized AOP conditions led to 78% Y, 82.2% RC, 42.7% DE, and 88.9% EC (overall glucose yield of 60.9%). Fermentation of the enzymatic hydrolysate with *Saccharomyces cerevisiae* yielded 190.8 L_{ethanol} ton_{SB}⁻¹. Biogas production by anaerobic digestion of residual liquid streams of the pretreatment steps yielded 27.46 NL_{CH₄} kg_{SB}⁻¹. An energy balance was estimated for the SB fractionation.

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1. Introduction

The growing global demand for energy and the resulting environmental impacts of an economy based on high use of fossil carbon sources are among the most important issues of the 21st century (IPCC, 2015). By 2050, global energy demand will be from 2 to 3 times higher than currently, making it unlikely that the existing fossil fuel reserves will be able to meet market demand (Sharma & Ghoshal, 2015). Lignocellulose biomass (LB) is a promising alternative to fossil fuels, providing cleaner and more sustainable energy, with high productivity and chemical composition suitable for the potential production of high added-value products. The conversion of LB into energy, materials, and fine chemicals can be performed in a lignocellulose feedstock (LCF) biorefinery, which is an industrial complex for the integral exploitation of LB (Diep et al., 2012).

Due to the presence of lignin, which has a structural function in plant cells, each type of LB has specific characteristics of recalcitrance and resistance to saccharification, making it necessary to use a pretreatment step to make the material more accessible for enzymatic hydrolysis (Rabelo et al., 2014). To this end, a highly efficient LB fractionation strategy should be able to maximize the production of fermentable sugars, while minimizing the production of fermentation inhibitors. In addition, it is essential to keep investments in equipment and chemicals as low as possible, as well as to minimize water consumption (Meighan et al., 2017; Vallejos et al., 2012).

The hydrothermal pretreatment (HP) by autohydrolysis of sugarcane bagasse (SB) has the main advantage of providing efficient extraction of hemicelluloses, resulting in a liquor rich in C5 sugars and xylooligomers (XOS) (Baêta et al., 2016a; Vallejos et al., 2012). After hydrolysis of the XOS to produce C5 sugars, the C5 sugars syrup can be converted into second-generation (2G) ethanol by fermentation using suitable microorganisms. However, there are still difficulties associated with the application of this strategy on an industrial scale, making the production of biogas by anaerobic digestion (AD) an interesting alternative from the point of view of the energy balance for an LCF biorefinery. This process may be particularly attractive when biogas is produced from the liquid waste streams generated during the fractionation process (Baêta et al., 2016a; Dias et al., 2012).

HP of SB results in a low degree of delignification, so the pretreated material has high lignin content (Baêta et al., 2016a; Meighan et al., 2017). In addition, the residual lignin from HP is highly condensed and chemically modified with furans, which leads to low accessibility and unproductive adsorption of enzymes (Carvalho et al., 2015). This increases the costs in the enzymatic hydrolysis step, because higher enzymatic loadings are required to achieve high conversion of cellulose to glucose. Therefore, an additional pretreatment step may be required to remove the residual lignin from the pretreated material, as its removal is difficult using conventional pretreatments such as acid or base-catalyzed and organosolv processes (Meighan et al., 2017).

Advanced oxidative pretreatment (AOP) can cleave the carbon-carbon bonds in condensed lignin, increasing its solubility in water (Gould, 1984). Therefore, AOP should be investigated as an option for further delignification of LB pretreated by HP. The application of AOP to LB that has not been pretreated requires only low reaction temperature (< 60 °C) and pressure (1 atm) and has the advantages that it produces a pretreated material suitable for achieving a high enzymatic conversion (> 90%), without generating fermentation inhibitors (Rabelo et al., 2014; Su et al., 2015; Xue et al., 2016; Yuan et al., 2018). However, drawbacks of this technology are the high amounts of H₂O₂ and NaOH used, as well as significant water consumption (Rabelo et al., 2014; Xue et al., 2016).

Therefore, the aim of this study was to evaluate the fractionation of SB by HP followed by alkaline extraction (AE) and AOP, for the production of 2G ethanol and biogas. The main bottlenecks of AOP (concerning the need to reduce the amounts of H₂O₂, NaOH, and water consumed) were investigated, with optimization of these parameters

using a Doehlert experimental design (DED). The best fractionation condition was used for 2G ethanol production by fermentation. In addition, the residual liquid streams of the SB fractionation were used for biogas production, in order to recover energy and also to treat the effluents from the fractionation process.

2. Material and methods

2.1. Chemicals

Sulfuric acid (95–98 wt%), sodium hydroxide (pellets), hydrogen peroxide (35 wt% in H₂O), cyclohexane, ethanol (99.5%), and anhydrous citric acid were purchased from Neon (Brazil). Sodium bicarbonate (≥ 99%) was purchased from Nuclear (Brazil). Hydrochloric acid (37 wt% in H₂O) was purchased from Alphatec (Brazil). Sulfuric acid (99.999%) was purchased from Sigma-Aldrich. Chromatography grade standards of cellobiose (≥ 98%), D-glucose, D-xylose, L-arabinose (≥ 99.5%), acetic acid (AA, 49–51 wt% in H₂O), formic acid (FA, 49–51 wt%), 5-hydroxymethyl-2-furfuraldehyde (HMF, ≥ 98.0%), 2-furfuraldehyde (FF, ≥ 98.5%), and ethanol (≥ 99.5%) were purchased from Sigma-Aldrich. Sodium azide was purchased from Vetec (Brazil). The enzymatic cocktails Cellic HTec2® and Cellic CTec2® were kindly provided by Novozymes (Araucária, Paraná, Brazil).

2.2. Sugarcane bagasse preparation

Raw SB, collected during the 2016/2017 harvest, was kindly provided by the Jatiboca Sugar and Ethanol Plant (Urucânia, Minas Gerais, Brazil). The raw SB containing ~50% moisture was dried under sunlight for 15 days, until ~10% equilibrium moisture content was achieved, followed by storage in a closed container, at room temperature (~22 °C).

2.3. Hydrothermal pretreatment by autohydrolysis of sugarcane bagasse

HP of the SB was performed using a floor-standing pressure reactor (Model 4532, Parr Instruments) equipped with a 1.9 L volume C-276 alloy vessel and a reactor controller (Model 4848, Parr Instruments). For all experiments, the reactor vessel was loaded with 120 g (dry weight basis) of raw SB. The HP conditions were those optimized and reported by Baêta et al. (2016a): 183 °C, 41 min, and liquid-to-solid ratio (LSR) of 3.94 mL g⁻¹. The heating time required to reach the pretreatment temperature was 41 min, after which the reactor vessel was kept at 183 °C for 17 min, in order to obtain a severity factor (S₀) of 3.78. The reactor pressure at 183 °C was 11.7 bar. At the end of the pretreatment time, the heating was stopped and the reactor vessel was quenched in an ice-water bath and slowly depressurized. The pretreated SB was pressed in a hydraulic press (Model SL10/15/E, SOLAB) at 9 tons pressure, in order to extract the hemicelluloses hydrolysate (HH). The HH and the pretreated SB cake were stored at -20 °C and were subsequently characterized to determine their chemical compositions (Section 2.9).

A graph of temperature against time was constructed for the heating, isothermal, and cooling phases of the reactor, in order to calculate the S₀ value (Overend et al., 1987) (Eq. (1)) of the HP, using data reported by Baêta et al. (2016a):

$$\ln S_0 = \int_0^t e^{\left(\frac{T-100}{14.75}\right)} dt \quad (1)$$

where t (min) is the time and T (°C) is the operating temperature.

2.4. Alkaline extraction of pretreated SB

A step consisting of AE of the SB pretreated by HP was applied to remove the lignin fraction partially modified in the HP, which was not soluble in the acidic aqueous phase (pH ~ 3.5–4.0). For this, SB pretreated by HP was dried in a forced-air oven (Model SL-102, SOLAB) at

55 °C for 12 h, followed by grinding in a Wiley mill (Model MA058, Marconi), in order to pass through a 40-mesh (0.42 mm) sieve. The AE was performed in a blunt-bladed blender loaded with an aqueous 0.2 mol L⁻¹ NaOH solution, at LSR of 10 mL g⁻¹, using a speed of 2,800 rpm for 30 s. The solid fraction was then washed with tap water on a 200-mesh (0.074 mm) screen, until neutral pH was reached. The black liquor (effluent) was stored at -20 °C for subsequent biogas production by AD (Section 2.8). The alkaline-extracted pretreated SB was stored at -20 °C, prior to characterization to determine its chemical composition (Section 2.9).

2.5. Advanced oxidative pretreatment of sugarcane bagasse

The AOP was performed in a white acrylic cylindrical 2 L reactor equipped with a mechanical stirrer with a three-disc acrylic shaft with grooves, in order to improve the mass transfer during the pretreatment. In a typical run, the reactor vessel was loaded with 30 g (dry weight basis) of alkaline-extracted pretreated SB, followed by the addition of distilled water, an aqueous H₂O₂ solution, and drops of an aqueous 5 mol L⁻¹ NaOH solution, to give a defined LSR and to adjust the solution pH to 11.6 (Gould, 1984). After the solution pH had been adjusted to 11.6, the suspension was stirred at 150 rpm and 25 °C for a defined time interval. During the AOP, a pH value of ~11.6 was maintained by adding drops of an aqueous 5 mol L⁻¹ NaOH solution. The composition of each reaction medium, in terms of applied H₂O₂ load and LSR, was obtained from the DED (Section 2.11). At the end of the AOP, the solid and liquid fractions were transferred to a blunt-bladed blender, operated at 2,800 rpm for 30 s, in order to deconstruct the cellulose pulp fibers. The cellulose pulp was then washed with tap water on a 200-mesh (0.074 mm) screen, until neutral pH was reached. The liquid fraction (effluent) was stored at -20 °C, prior to biogas production by AD (Section 2.8). The wet solid fraction (~70% moisture content) was stored at -20 °C, prior to subsequent determination of its chemical composition (Section 2.9) and use in the enzymatic hydrolysis experiments (Section 2.6). The residual cellulose content (RC) and degree of delignification (DE) after the AOP were calculated as shown in Eqs. (2) and (3), respectively (Novo et al., 2011):

$$RC/\% = \left[\frac{C^{RSF}(Y^{RSF}/100)}{C^{\text{pretreated SB}}} \right] \times 100 \quad (2)$$

where $C^{\text{pretreated SB}}$ (%) and C^{RSF} (%) are the cellulose contents in the alkaline-extracted pretreated SB and in the residual solid fraction (RSF) from the AOP, respectively, and Y^{RSF} (%) is the yield (the amount of RSF obtained after AOP of the alkaline-extracted pretreated SB).

$$DE/\% = \left\{ \frac{L^{\text{pretreated SB}} - [L^{RSF}(Y^{RSF}/100)]}{L^{\text{pretreated SB}}} \right\} \times 100 \quad (3)$$

where $L^{\text{pretreated SB}}$ (%) and L^{RSF} (%) are the lignin contents of the alkaline-extracted pretreated SB and the residual solid fraction after the AOP, respectively.

2.6. Enzymatic hydrolysis of pretreated sugarcane bagasse

Enzymatic hydrolysis experiments were performed in 25 mL Erlenmeyer flasks containing 1.000 g (dry weight basis) of RSF from the AOP and 10 mL of a mixture consisting of an aqueous 0.05 mol L⁻¹ citrate buffer solution (pH 4.8) with 0.02% (m v⁻¹) sodium azide and a mixture of enzymatic cocktails (15% Cellic HTec2® and 85% Cellic CTec2®), giving an LSR of 10 mL g⁻¹. The flasks were incubated for 72 h in an orbital shaker incubator (Model MA-830, Marconi), at 150 rpm and 50 °C. Cellulase activity was determined in terms of filter paper unit (FPU) (Ghose, 1987). The enzymatic activity was 264.29 FPU mL⁻¹. The loads of cellulase and β-glucosidase were 10 FPU and 52.2 IU (international units) per gram of solid fraction, respectively.

At the end of the hydrolysis, the flasks were immersed in an ice-water bath to stop the hydrolysis reaction and were then centrifuged at 4500 rpm for 15 min to separate the solid and liquid fractions. The liquid fraction rich in sugars was characterized to determine the concentrations of sugars, sugar degradation products, and organic acids (Section 2.9).

The enzymatic conversion (EC) of cellulose into glucose was calculated using Eq. (4):

$$EC/\% = \left[\frac{w_{\text{glucose}} f_h}{w_{\text{substrate}} (C^{\text{substrate}}/100)} \right] \times 100 \quad (4)$$

where w_{glucose} (g) is the glucose content in the hydrolysate obtained after enzymatic hydrolysis and $w_{\text{substrate}}$ (g) is the weight of the solid fraction used in the enzymatic hydrolysis, f_h (0.9) is the glucose-to-cellulose conversion factor, and $C^{\text{substrate}}$ (%) is the cellulose content in the substrate (solid fraction).

The overall glucose yield (OGY) was calculated using Eq. (5):

$$OGY/\% = \left[\frac{w_{\text{glucose}}}{w_{\text{raw SB}} (C^{\text{raw SB}}/100)(1/f)} \right] \times 100 \quad (5)$$

where $w_{\text{raw SB}}$ (g) is the weight of raw SB used at the beginning of fractionation process and $C^{\text{raw SB}}$ (%) is the cellulose content in the raw SB.

2.7. Second-generation ethanol production by fermentation

The best AOP conditions (0.3 g_{H2O2} g_{SB}⁻¹, 9.67 mL g_{SB}⁻¹, 8 h) were used to produce a sugar-rich liquor by enzymatic hydrolysis, for subsequent fermentation into 2G ethanol. Three strains of *Saccharomyces cerevisiae* yeast were used in the fermentation experiments, including two commercial yeast strains used in 1G ethanol production in Brazilian mills (PE-2 and CAT-1, Fermentec, Brazil) and one yeast strain (LBCM1047) isolated from Brazilian *cachaça* distilleries (Araujo et al., 2018). The fermentations were carried out in cylindrical glass bottles (80 mL) containing 45 mL of a sugar-rich liquor (2.02 g L⁻¹ cellobiose, 78.73 g L⁻¹ glucose, 5.75 g L⁻¹ xylose, 0.51 g L⁻¹ arabinose, and 0.26 g L⁻¹ formic acid), previously autoclaved at 120 °C for 20 min. The flasks were placed in an incubator at 34 °C, with stirring at 150 rpm. The inoculation was made with 4×10^7 cells mL⁻¹, corresponding to an optical density at 600 nm (OD₆₀₀) of 4.0. The experiments were performed in triplicate and a control experiment without substrate addition was also performed. No nutrient solution was added to the glass bottles. The bottles were periodically weighed during the fermentation, in order to record the weight loss due to CO₂ production and follow the progress of the fermentation. At the end of the fermentation, the fermented medium was withdrawn and centrifuged (12,000 rpm, 20 min, 25 °C). The chemical composition of the supernatant was then determined by high performance liquid chromatography (HPLC) (Section 2.9).

2.8. Biogas production by anaerobic digestion

The effluents from the AE and AOP pretreatment steps were tested separately for biogas production by AD in batch assays. The experiments were performed in 120 mL glass bottles, under mesophilic conditions. The flasks were loaded with 60 mL volumes of AE or AOP effluent samples, together with the inoculum. The amount of the latter depended on the chemical oxygen demand (COD) of each effluent and was sufficient to obtain a food-to-microorganism (F/M) ratio of 0.375 g_{COD} g_{VSS}⁻¹ in each test. The COD was determined according to Standard Method 5220D (Chemical Oxygen Demand). An aqueous sodium bicarbonate buffer solution was used to keep the suspension pH at 7.0. A solution containing micro and macronutrients was also added to each flask, in order to maintain a minimum COD:N:P ratio of 350:5:1, as reported by Baêta et al. (2013). The total volume of the suspension was 60 mL, allowing a headspace of 60 mL during the tests. The flasks were incubated in an orbital shaker incubator (Model SL 221, Solab), at 200 rpm and 35 °C. The biogas production was determined by

measuring the accumulated pressure in the bottles, using a manometer (Model PM-9100HA, Lutron). The composition of the biogas was determined by gas chromatography (Section 2.9). In all the experiments, a control assay without substrate was used to account for the endogenous methanogenic activity of the inoculum. Biochemical methane potential (BMP) tests were stopped when the daily methane production was less than 5% of the total accumulated production. All the BMP values (in $\text{Nm}^3_{\text{CH}_4} \text{kg}_{\text{COD}}^{-1}$) were expressed using standard temperature and pressure conditions (273 K, 101,315 Pa), as defined by the International Union of Pure and Applied Chemistry (IUPAC). All the experiments were performed in duplicate and the results were averaged.

2.9. Estimation of the energy balance for sugarcane bagasse fractionation

Estimation of the energy balance for SB fractionation was performed considering the amount of energy consumed in the pretreatment steps prior to enzymatic hydrolysis, compared to the amount of energy generated by combustion of the biogas produced by AD in a combined heat and power (CHP) system. Considering the typical efficiency values of a commercial CHP system, 10% of the energy generated by the biogas combustion is lost, with 33.3% of the remainder converted into electrical energy (E_E) and 66.6% converted into potential thermal energy (E_T) (Pérez-Elvira & Fdz-Polanco, 2012).

The energy consumed in the HP of the SB was estimated from the energy required to heat the inlet SB from 25 °C to the reaction temperature (183 °C), and to heat the water (LSR of 3.94 L kg^{-1}) from 100 °C to 183 °C, as shown in Eq. (6):

$$E_{T(-),HP}/MJ = [C_{p,SB} w_{SB} (T_{HP,T} - T_{SB,25^\circ\text{C}})] + [C_{p,H_2O} \rho_{H_2O} LSR w_{SB} (T_{HP,T} - T_{H_2O,100^\circ\text{C}})] \quad (6)$$

where $E_{T(-),HP}$ (MJ) is the thermal energy consumed in the HP of the SB, $C_{p,SB}$ and C_{p,H_2O} are the specific heat capacities of SB ($0.00176 \text{ MJ (kg } ^\circ\text{C)}^{-1}$) and water ($0.00419 \text{ MJ (kg } ^\circ\text{C)}^{-1}$) (Kling et al., 1987), respectively, w_{SB} (kg) is the weight of SB, $T_{HP,T}$ is the HP temperature (183 °C), $T_{SB,25^\circ\text{C}}$ and $T_{H_2O,100^\circ\text{C}}$ (°C) are the inlet temperatures of SB (25 °C) and water (100 °C), respectively, LSR (L kg^{-1}) is the liquid-to-solid ratio, and ρ_{H_2O} (kg L^{-1}) is the water specific gravity. The inlet water temperature of 100 °C was based on the use of water from the condensation turbines of the CHP system.

The SB pretreated by HP was prepared for the AE step by drying in a forced-air circulation oven, so that the moisture content decreased from ~50% to 15%. After drying, the particle size of the material was reduced by grinding in a Wiley mill. The energy consumed in the drying process was estimated from the amount of water heated and vaporized in this step, as shown in Eq. (7):

$$E_{T(-),drying}/MJ = w_{H_2O} [C_p (\Delta T) + Q] \quad (7)$$

where $E_{T(-),drying}$ (MJ) is the energy consumed in the drying of the SB pretreated by HP and Q (2.256 MJ kg^{-1}) is the latent heat for water vaporization.

The energy consumed for milling the SB pretreated by HP was calculated according to the Bond law, as shown in Eq. (8):

$$E_{E(-),milling}/kWh = w_i C \left(\frac{1}{\sqrt{D_2}} - \frac{1}{\sqrt{D_1}} \right) \quad (8)$$

where $E_{E(-),milling}$ (kWh) is the energy consumed in the milling of SB pretreated by HP, w_i (kWh kg^{-1}) is the work index of the material to be milled, C is the amount of HP-pretreated SB fed into the Wiley mill (kg), D_2 is the desired average particle size (0.42 mm), and D_1 is the average particle size at the inlet (11.0 mm).

For the AE and AOP steps, the energy consumed was estimated using Eq. (9):

$$E_{E(-),AE,AOP}/kWh = (P t_{op}) 0.278 \quad (9)$$

where $E_{E(-),AE,AOP}$ (kWh) is the energy consumed in the AE and AOP steps, P is the power of the equipment used (650 W for AE and 150 W for AOP), t_{op} (h) is the operating time of the equipment, and 0.278 is the factor for conversion to kWh.

Finally, the thermal energy generated by the combustion of CH_4 produced from AD of the liquid waste stream in a CHP system (Eq. (10)) was subtracted from the total thermal energy consumed in the HP and the drying of SB pretreated by HP. The electric energy produced (Eq. (11)) was subtracted from the total electric energy consumed in the pretreatment steps for fractionation of the SB. The energy balance for the SB fractionation is then given by Eqs. (10)–(12), as follows:

$$E_{T(+),MJ} = BMP \left(\frac{\text{Nm}^3 \text{CH}_4}{\text{kg COD}} \right) \left(\frac{\text{kg COD}}{\text{kg SB}} \right) C_{p,\text{CH}_4} (0.9)(0.66) \quad (10)$$

$$E_{E(+),kWh} = BMP \left(\frac{\text{Nm}^3 \text{CH}_4}{\text{kg COD}} \right) \left(\frac{\text{kg COD}}{\text{kg SB}} \right) C_{p,\text{CH}_4} (0.9)(0.33)(0.278) \quad (11)$$

$$\Delta E_{T,E} = E_{T,E}(+) - E_{T,E}(-) \quad (12)$$

where BMP ($\text{Nm}^3_{\text{CH}_4} \text{kg}_{\text{soluble COD,AE/AOP}}^{-1}$) is the biochemical methane potential of the liquid waste stream generated from both AE and AOP of the pretreated SB, C_{p,CH_4} ($34.45 \text{ MJ (kg } ^\circ\text{C)}^{-1}$) is the methane specific heat capacity, 0.9 is the biogas combustion efficiency, and 0.66 and 0.33 are the yields of thermal and electric energies produced, respectively, considering a CHP system fed with biogas.

2.10. Analytical methods used for characterization of the solid, liquid, and gas fractions

The chemical compositions of the raw SB and the solid fractions recovered after each pretreatment step were determined according to TAPPI (Technical Association of the Pulp and Paper Industry) and NREL (National Renewable Energy Laboratory) standard methods for extractives (cyclohexane was used instead of benzene) (TAPPI, 1997), cellulose and hemicelluloses (Ruiz & Ehrman, 1996), acid-soluble lignin (Golschmidt, 1971), acid-insoluble lignin (Templeton & Ehrman, 1995), and ash (TAPPI, 1993). Each test for determination of the chemical compositions of the solid fractions was performed at least in triplicate and the results were averaged.

Separation and quantification of the C5 and C6 sugars, sugar degradation products, organic acids, and ethanol was performed by high performance liquid chromatography (HPLC), using an Agilent Technologies 1260 instrument equipped with a microguard cartridge (Cation H^+ , Aminex® HPX-87H, Bio-Rad) and an Aminex® HPX-87H column ($300 \times 7.8 \text{ mm}$, Bio-Rad) kept at 55 °C. The instrument was operated in isocratic elution mode, using aqueous $5 \text{ mmol L}^{-1} \text{H}_2\text{SO}_4$ solution as the eluent, at a flow rate of 0.6 mL min^{-1} . A refractive index detector was used for detection of the C5 and C6 sugars and ethanol. The organic acids and sugar degradation products were detected using a UV-Vis detector, at wavelengths of 210 and 274 nm, respectively.

For mass balances, the analytes separated and quantified by HPLC analysis were converted into amounts of cellulose and hemicelluloses using conversion factors, as shown in Eqs. (13) and (14):

$$\text{Cellulose}/(\%) = 0.95\text{Cel}_{\%} + 0.90\text{Glu}_{\%} + 1.286\text{HMF}_{\%} + 3.522\text{FA}_{\%} \quad (13)$$

$$\text{Hemicelluloses}/(\%) = 0.88\text{Xyl}_{\%} + 0.88\text{Ara}_{\%} + 1.375\text{FF}_{\%} + 0.717\text{AA}_{\%} \quad (14)$$

where $\text{Cel}_{\%}$, $\text{Glu}_{\%}$, $\text{HMF}_{\%}$, $\text{FA}_{\%}$, $\text{Xyl}_{\%}$, $\text{Ara}_{\%}$, $\text{FF}_{\%}$, and $\text{AA}_{\%}$ are the percentages (w w^{-1}) of cellobiose, glucose, HMF, FA, xylose, arabinose, FF, and AA, respectively.

Determination of the biogas composition, in terms of methane (v v^{-1}), employed a gas chromatograph (GC) (Model 2014/TCD, Shimadzu) equipped with a thermal conductivity detector and a capillary column ($30 \text{ m} \times 0.53 \text{ mm}$, Restek) packed with molecular sieve

5 Å and kept at 120 °C. The detector was operated at a current of 85 mA and temperature of 120 °C. The carrier gas was nitrogen, at a flow rate of 1.66 mL min⁻¹. An analytical curve constructed using a standard gas mixture (24.99 mol_{CH4} mol_{mixture}⁻¹) was employed for quantification of methane.

2.11. Design of experiments and statistical analysis

A DED was created with 12 assays and triplicate at the center point. This design was used to evaluate the AOP conditions by varying three independent variables at different levels. These were the ratio of hydrogen peroxide to alkaline extracted pretreated SB (from 0.3 to 3.0 g_{H2O2} g_{SB}⁻¹, 7 levels), *LSR* (from 5 to 25 mL g⁻¹, 5 levels), and reaction time (from 1 to 8 h, 3 levels). The dependent variables (*DV*) optimized were *Y*, *RC*, *DE*, and *EC*.

The experimental data were evaluated using Statistica® 12.0 (StatSoft, Inc.) routines for analysis of variance (ANOVA), graphical analysis (Pareto charts and response surfaces), modeling of the responses (*DVs*), and regression coefficients. The pure error was selected for statistical significance and computation of experimental errors. The statistical analyses were carried out with a confidence level of 95%. The most complex model, including interactions between the independent variables, was used to model the responses. The second-degree polynomial model used to fit the experimental data is shown in Eq. (15):

$$DV = a_0 + a_1 LSR + a_2 LSR^2 + a_3 g_{H_2O_2} g_{SB}^{-1} + a_4 g_{H_2O_2} g_{SB}^{-2} + a_5 t + a_6 t^2 + a_7 LSR \times g_{H_2O_2} g_{SB}^{-1} + a_8 LSR \times t + a_9 g_{H_2O_2} g_{SB}^{-1} \times t \quad (15)$$

where $a_0, a_1, a_2, \dots, a_9$ are regression coefficients calculated by fitting of the second-degree polynomial model to the experimental data.

2.12. Scanning electron microscopy

Samples of RSF from the AOP were dried in a forced-air circulation oven (Model SL-102, SOLAB) at 60 °C for 3 days. Samples were not milled after drying to preserve their structure. Dried samples were dispersed on aluminum sample holders with a conductive double-sided tape of carbon. Samples were sputter-coater with carbon and gold in a high-vacuum coater (Model Q150R ES, Quorum technologies) and examined on a scanning electron microscope (Model Vega3 SB, Tescan/Oxford Instruments) operating with a tungsten filament (10–20 keV) and a detector of secondary electrons.

3. Results and discussion

3.1. Characterization of the raw sugarcane bagasse

The chemical composition of the raw SB used in this study is presented in Table 1. Rocha et al. (2015) characterized different sugarcane varieties cultivated in Brazil and reported an average composition of 42.10 ± 1.93% cellulose, 27.60 ± 0.88% hemicelluloses, 21.56 ± 1.67% lignin, 5.63 ± 2.31% extractives, and 2.84 ± 1.22% ash. Differences between the compositions of the raw SB used in this study and in previous work (Rocha et al., 2015) were related to soil conditions, climate, variety, maturation stage, and harvest season.

3.2. Hydrothermal pretreatment by autohydrolysis of SB followed by alkaline extraction

The raw SB was pretreated by HP using the experimental conditions described by Baêta et al. (2016a): 183 °C, 41 min, and *LSR* of 3.94 mL g⁻¹. Table 1 shows the chemical compositions of the raw SB and the solid fraction of the raw SB pretreated by HP, in terms of cellulose, hemicelluloses, and lignin. The HP of the SB led to a weight loss of 29.5%, hemicelluloses removal of 73%, and cellulose preservation of 82.1%. This high level of hemicelluloses removal, combined with high

Table 1

Chemical compositions and yields for the pretreatments applied to sugarcane bagasse.

Parameter/(%)	Raw SB	HP-pretreated SB	AE of HP-pretreated SB
Cellulose	48.70 ± 0.50	56.74 ± 1.74	78.91 ± 1.42
Hemicelluloses	21.14 ± 0.21	8.1 ± 0.47	7.13 ± 0.57
Lignin	24.81 ± 0.44	33.5 ± 2.31	17.6 ± 1.09
Extractives	2.31 ± 0.20	–	–
Ash	0.41 ± 0.03	2.68 ± 0.11	1.55 ± 0.16
Yield (<i>Y</i>)	–	70.5	74.5
Weight loss	–	29.5	25.5
Residual cellulose (<i>RC</i>)	–	82.14 ± 3.4	103.61 ± 3.32
Hemicelluloses removal (<i>HR</i>)	–	72.99 ± 2.03	34.42 ± 7.15
Delignification (<i>DE</i>)	–	6.09 ± 1.81	61.75 ± 3.79
Enzymatic conversion (<i>EC</i>)	14.86	34.55 ± 2.52	46.82 ± 2.05
Overall glucose yield (<i>OGY</i>)	14.86	29.20	38.46
IL-to-SL ratio	8.93	11.85	26.81

cellulose preservation, led to HH with a high concentration of monosaccharides and a low concentration of sugar degradation products (20.87 ± 0.17 g L⁻¹ *XOS*, 2.65 ± 0.02 g L⁻¹ glucose, 15.31 ± 0.34 g L⁻¹ xylose, 0.86 ± 0.09 g L⁻¹ arabinose, 10.49 ± 0.26 g L⁻¹ formic acid, 11.92 ± 0.24 g L⁻¹ acetic acid, 0.58 ± 0.00 g L⁻¹ *HMF*, and 1.33 ± 0.01 g L⁻¹ *FF*).

The use of HP for the pretreatment of SB is economically attractive, as demonstrated by Baêta et al. (2016b), who showed that the energy required for this pretreatment could be produced by two-stage (acidogenic followed by methanogenic) AD of HH, leading to a net energy of 3.15 MJ kg_{SB}⁻¹. In addition, it uses only water as the solvent and reagent for the hydrolysis reactions, without the need for addition of chemicals. In this pretreatment method also known as autohydrolysis, an increase in the reaction temperature (from 25 to 183 °C) leads to a decrease in the *pK_w* of water, consequently increasing the concentration of hydronium ions (H₃O⁺) in the reaction medium (Meighan et al., 2017). The H₃O⁺ ions are responsible for catalyzing the hydrolysis reactions of the acetyl groups of the hemicelluloses, resulting in the release of acetic acid (11.92 g L⁻¹; *pK_a* of 4.75) and an additional reduction of the pH of the reaction medium. This additional acidity accelerates the hydrolysis reactions of the ether bonds of the hemicelluloses, releasing soluble *XOS* (20.87 g L⁻¹) to the reaction medium (Meighan et al., 2017), which are subsequently converted into xylose. Due to the high temperature of HP, fractions of the xylose and arabinose are further converted into *FF* and formic acid (10.49 g L⁻¹, *pK_a* of 3.77) (Ruiz et al., 2017). The generation of formic acid in the reaction medium also leads to a decrease of the pH. The lower acidity of the HP reaction medium, compared to acid-catalyzed pretreatments with strong acids, results in substantially lower degradation of monosaccharides to produce furans (*FF* and *HMF*) (Baêta et al., 2016a).

After HP of the SB, the lignin content increased from 24.8 to 33.5%, although a *DE* value of 6.1% was obtained. During HP, cleavage of the bonds between lignin and hemicelluloses is expected to occur. The cleavage of α- and β-O-4 aryl and alkyl bonds in lignin can also occur, leading to the release into the reaction medium of lower molar mass lignin fragments with higher reactivity (Leschinsky et al., 2008a, 2008b). These low molar mass fragments have low solubility in the acidic reaction medium and high toxicity to microorganisms used for the production of 2G ethanol and biogas (Baêta et al., 2016a; Girio et al., 2010; Vallejos et al., 2012). In addition, lignin fragments within the plant cell can be repolymerized and/or condensed with carbohydrates and/or furans, promoting the formation of new carbon–carbon (C–C) bonds between lignin units, such as β-β, β-1, and β-5 bonds, which also leads to the formation of chemically modified lignin (Carvalho et al., 2015; Leschinsky et al., 2008a, 2008b).

The formation of chemically modified lignin may be observed from the ratio of insoluble to soluble lignin (IL-to-SL) in the pretreated SB. It was evident that the lignin of the SB was chemically modified during

HP, because the IL-to-SL ratio increased from 8.93 to 11.85. This also suggested that the modified lignin could be more difficult to remove using delignification methods based on the cleavage of ether bonds, such as the soda, kraft, and organosolv (acid or base catalyzed) techniques, as reported by [Carvalho et al. \(2015\)](#) and [Meighan et al. \(2017\)](#).

In addition, HP carried out at low LSR, as in the present work, can result in higher acidity of the reaction medium, consequently leading to increases of the cleavage of ether bonds and the number of lignin condensation reactions. Therefore, the SB pretreated by HP was submitted to a washing step using aqueous 0.2 mol L^{-1} NaOH solution. The purpose of this additional step was to evaluate the efficiency of removal of low molar mass lignin fragments from the SB pretreated by HP. These fragments may be deposited on the surfaces of the fibers and/or be present within the plant cells (without chemical bonding). As shown in [Table 1](#), a DE value of 61.8% was obtained after AE of the SB pretreated by HP, indicating that a substantial amount of the lignin that was not chemically bonded to the lignocellulose matrix could be extracted by means of a simple washing step. The IL-to-SL ratio increased from 11.85 to 26.81 after AE, indicating that after AE, the pretreated SB contained a much greater amount of chemically modified residual lignin.

Evaluation was made of the effect on the enzymatic hydrolysis caused by the low molar mass lignin that could be extracted (by AE) from the SB pretreated by HP. The results obtained for enzymatic hydrolysis of SB pretreated by HP, before and after AE ([Table 1](#)), showed that the EC value increased by 33% (from 34.6 to 46.8%) (OGY value increased by 32% from 29.2 to 38.5%), indicating that the lignin extracted by AE, which was not chemically bonded to the lignocellulose matrix, contributed negatively to the enzymatic hydrolysis. However, the low EC value for SB pretreated by HP after AE suggested that the chemically modified residual lignin could have a fundamental influence on the low EC value of this material. Although the EC value of SB pretreated by HP increased significantly after AE, the EC value of 46.8% (OGY value of 38.5%) was not sufficiently high to allow the design of an economically viable integrated and/or stand-alone 2G ethanol production plant, according to the scenarios simulated by [Dias et al. \(2012\)](#). Therefore, a further pretreatment step was required in order to increase the access of the hydrolytic enzymes to the cellulose and reduce the costs associated with the enzymes, so as to improve the EC value of SB pretreated by HP at low LSR. In addition, the lignin extracted by AE could consume more H_2O_2 in the AOP, hence increasing the cost of the SB fractionation strategy proposed in this study. This justified the need for an AE step before the AOP.

3.3. Advanced oxidative pretreatment of SB pretreated by HP

Enzymatic hydrolysis is favored when the pretreated material employed has a high cellulose content ($\geq 70\%$) and a low lignin content ($\leq 10\%$) ([Rabelo et al., 2014](#)). The DED methodology was used to optimize the AOP conditions (ratio of hydrogen peroxide to alkaline extracted pretreated SB, LSR, and reaction time) for delignification of the alkaline-extracted SB pretreated by HP, aiming at the production of a solid fraction that presented the highest possible RC and EC values, together with the lowest possible weight loss. The simultaneous optimization of all the AOP variables for maximization of the responses was performed using the Statistica® desirability tool.

[Table 2](#) presents the results obtained from the DED experiments for each of the response variables evaluated. The minimum and maximum levels evaluated for the independent variables led to a good variability in the results obtained for the response variables, with DE values from 19.1 to 95.4%, Y values from 33.7 to 94.7%, EC values based on the pretreated SB from 48.5 to 93.5%, and OGY values based on the raw SB used at the beginning of the fractionation process from 20.9 to 67.8%. [Xue et al. \(2016\)](#) reported an EC value of 98.7% and an OGY value of 83.2% using reaction conditions of $0.36 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, LSR of $10 \text{ mL g}_{\text{SB}}^{-1}$, 40°C , and 24 h, with $30 \text{ FPU g}_{\text{pretreated SB}}^{-1}$ in the enzymatic hydrolysis step. In the present study, EC values of 92.4% (OGY of

30.8%) and 93.5% (OGY of 39.3%) were obtained using $1.65 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, LSR of $5 \text{ mL g}_{\text{SB}}^{-1}$, 25°C , and 4.5 h (experiment 4), and $0.3 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, LSR of $20 \text{ mL g}_{\text{SB}}^{-1}$, 25°C , and 4.5 h (experiment 7), respectively, with an enzymatic loading of $10 \text{ FPU g}_{\text{pretreated SB}}^{-1}$.

One of the advantages of the AOP conditions applied in the present study was the use of a low temperature (25°C), resulting in a lower energy demand, compared to conditions where heating of the reactor was required ([Xue et al., 2016](#); [Yuan et al., 2018](#)). In addition, the shorter reaction times led to lower energy consumption for agitation, as well as higher productivity. Compared to the study of [Rabelo et al. \(2014\)](#), who reported EC of 99% and OGY of 92.6% using $2.68 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, LSR of $25 \text{ mL g}_{\text{SB}}^{-1}$, 25°C , and 1 h, with 3.5 FPU g^{-1} and 3% of solids (LSR of 33.3 mL g^{-1}), the present work required lower amounts of H_2O_2 and water in the SB pretreatment, due to the high lignin removal efficiency in the AE step ($\sim 62\%$). This lignin had probably been depolymerized by cleavage of the α - and β -alkyl aryl ether bonds, but had not been removed from the SB pretreated by HP, due to its low solubility. Therefore, the oxidative reactions during the AOP involved the lignin fraction of the pretreated SB that could not be extracted by the AE, leading to cleavage of the C–C bonds formed by lignin repolymerization during HP of the SB.

The DE values observed in the present study can be compared to the higher DE values reported in the literature, such as DE of 92% reported by [Vallejos et al. \(2015\)](#), who investigated the organosolv pretreatment of SB with ethanol:water (1:1, v v⁻¹) at 190°C , for 150 min, using LSR of 3 g g^{-1} . [Rabelo et al. \(2014\)](#) reported DE of 87.1% using $2.68 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, LSR of $25 \text{ mL g}_{\text{SB}}^{-1}$, 25°C , and 1 h. Despite this, some high weight loss values were reported by [Rabelo et al. \(2014\)](#). Consequently, for those experimental conditions with high weight losses, the RC values were low, which negatively affected the overall yield of the 2G ethanol production process.

However, experiment 12 ([Table 2](#)) presented a high Y value (94.6%) and a high RC value (94.7%). Although the DE value was low (19.1%), high EC (87.1%) and OGY (67.8%) were obtained. This indicated that the extraction of a small fraction of the lignin (19.1%) by cleavage of the C–C bonds could significantly increase the enzymatic hydrolysis yield by allowing greater access of the cellulolytic enzymes to the substrate.

The Pareto charts for the standardized effects of the independent variables on the responses evaluated (Y, RC, DE, and EC) were used to evaluate the statistical significance of the variables. As expected, the applied H_2O_2 load variable had the greatest significant effect (8.310) on the DE of the SB pretreated by HP. This was in good agreement with the results reported in the literature for the use of other oxidative delignification agents, such as ozone ([Travaini et al., 2013](#)), or non-oxidative delignification agents, such as those employed in organosolv treatment ([Meighan et al., 2017](#); [Vallejos et al., 2015](#)). The *t* variable had a significant negative effect (-4.460) on DE, indicating that a longer reaction time could lead to the occurrence of undesired reactions that might hinder the removal of lignin from SB pretreated by HP. The models used to fit the responses presented R^2 values higher than 0.8. In addition, the models did not present any lack of fit (*p*-values > 0.05).

[Fig. 1a](#) shows the response surface for delignification of the alkaline-extracted SB pretreated by HP, as a function of reaction time and applied H_2O_2 load. For longer reaction times, only the use of a higher applied H_2O_2 load led to satisfactory DE values. For experimental conditions where lower applied H_2O_2 loads were combined with shorter reaction times, the radicals generated were probably rapidly consumed by the lignin and by the lignin oxidation products solubilized in the reaction medium. Hence, it was likely that during the remaining reaction time there were no oxidizing agents present at concentrations sufficient to cause additional removal of lignin from the pretreated SB ([Galindo-Hernández et al., 2018](#); [Gould, 1984](#)).

For the response RC, the only significant effect was the interaction between LSR and applied H_2O_2 load, which had a significant positive effect (4.676). For the response Y, LSR showed a significant positive effect (4.990), while applied H_2O_2 load had a significant negative effect

Table 2

Doehlert experimental design matrix and results obtained for AOP of alkaline extracted SB pretreated by HP.

Experiment	$\text{g}_{\text{H}_2\text{O}_2} \text{g}_{\text{SB}}^{-1}$	$\text{LSR}/(\text{mL g}^{-1})$	$t/(\text{h})$	$Y/(\%)$	$\text{RC}/(\%)$	$\text{DE}/(\%)$	$\text{EC}/(\%)$	$\text{OGY}/(\%)$
1	1.65	25	4.5	56.6	63.85	57.65	65.64	34.42
2	3.0	20	4.5	61.2	73.14	82.68	65.83	39.55
3	2.1	20	8	72.3	80.34	53.73	74.37	49.07
4	1.65	5	4.5	42.4	40.54	76.85	92.39	30.76
5	0.3	10	4.5	73.5	74.28	48.22	77.70	47.40
6	1.2	10	1	57.0	64.32	68.17	64.98	34.33
7	0.3	20	4.5	67.5	51.23	35.80	93.50	39.34
8	1.2	20	1	80.5	74.46	80.97	87.19	53.32
9	3.0	10	4.5	33.7	39.66	85.47	67.76	22.07
10	2.55	15	1	40.3	52.54	95.37	48.51	20.93
11	2.1	10	8	48.9	49.73	94.99	93.55	38.21
12	0.75	15	8	94.6	94.72	19.07	87.11	67.77
13C	1.65	15	4.5	57.6	65.40	70.44	67.50	36.26
14C	1.65	15	4.5	65.3	74.02	57.60	73.11	44.45
15C	1.65	15	4.5	56.3	62.37	68.97	72.07	36.92

(−7.823). In addition, the interaction between *LSR* and applied H_2O_2 load had a non-significant positive effect on *Y* (3.449). These results suggested that for the responses *RC* and *Y*, the effects of *LSR*, applied H_2O_2 load, and the interaction between *LSR* and applied H_2O_2 load were similar, because the higher weight losses observed after the AOP (Table 2) were probably due to greater solubilization of the carbohydrates, caused by high amounts of H_2O_2 available in the reaction medium. This demonstrated that an increase in the applied H_2O_2 load decreased the *Y* values, consequently increasing the amount of solubilized cellulose in the reaction medium.

Fig. 1b and c present the response surfaces for *RC* and *Y*, as a function of *LSR* and applied H_2O_2 load, respectively, with the reaction time fixed at 4.5 h. Comparing the two response surfaces presented in Fig. 1a and b, it can be seen that higher *Y* and *RC* values were obtained with lower applied H_2O_2 loads ($< 1.5 \text{ g}_{\text{H}_2\text{O}_2} \text{g}_{\text{pretreated SB}}^{-1}$) and medium *LSR* values (10–15 $\text{mL g}_{\text{pretreated SB}}^{-1}$), resulting in *RC* higher than 80%. In addition, increases of both *LSR* and applied H_2O_2 load led to increased *RC* in the solid fractions resulting from the AOP, but also led to lower *Y* values.

For the response *EC*, significant effects were found for the applied H_2O_2 load (−8.429) and the reaction time (7.438). This indicates that longer reaction times increased the *EC* values, as also reported by Xue et al. (2016). Differently, Rabelo et al. (2014) only observed a positive effect for the H_2O_2 concentration, while Xue et al. (2016) observed significant positive effects for both time and the applied H_2O_2 load. It should be noted that in these studies, there was no evaluation of the amount of water used in the pretreatments performed using design of experiments. Rabelo et al. (2014) and Xue et al. (2016) used *LSR* of 25 and 10 $\text{mL g}_{\text{SB}}^{-1}$, respectively, while in both cases the AOP was applied using H_2O_2 in alkaline medium with raw SB. In the present study, an AE step was applied for previous delignification of the SB pretreated by HP, since the structural changes that occurred in the lignin during HP of the SB made it more difficult to remove, as discussed previously (Section 3.2).

Therefore, the attempt to remove the lignin fraction closely associated with the cellulose fraction of alkaline-extracted SB pretreated by HP probably increases the possibility of carbohydrate solubilization (Rabelo et al., 2014). On the other hand, when cellulose solubilization occurs, it is most likely that the amorphous fraction is solubilized, because this fraction is more easily accessible to the radicals ($\text{HO}\cdot$ and $\cdot\text{O}_2^-$), leaving the higher crystallinity fraction in the pretreated SB. Obviously, the higher crystallinity cellulose fraction is also the fraction that presents greater difficulty of access for the hydrolytic enzymes (Ling et al., 2017).

Fig. 1d and e show the response surfaces for *EC*, as a function of applied H_2O_2 load and reaction time, with *LSR* fixed at 15 mL g^{-1} , and as a function of *LSR* and applied H_2O_2 load, with *t* fixed at 1 h, respectively. High *EC* values were achieved using low applied H_2O_2 loads ($< 1 \text{ g}_{\text{H}_2\text{O}_2} \text{g}_{\text{pretreated SB}}^{-1}$) and shorter reaction times. For the SB

fractionation strategy investigated in this study, it was not necessary to use higher amounts of delignification agent to obtain higher *EC* values. In addition, the combination of the variables *LSR* and applied H_2O_2 load at the higher levels negatively affected *EC*. For *LSR* fixed at 15 mL g^{-1} , it was necessary to use a lower applied H_2O_2 load, in order to avoid low *EC*. For time fixed at 1 h, the use of an applied H_2O_2 load of 0.5 $\text{g}_{\text{H}_2\text{O}_2} \text{g}_{\text{pretreated SB}}^{-1}$ only resulted in higher *EC* at *LSR* higher than 15 mL g^{-1} .

Therefore, it could be concluded that the amount of water used in the AOP had an important effect on the generation of a solid fraction suitable for enzymatic conversion. In addition, a certain dilution of the H_2O_2 in the reaction medium was required, in order to ensure that the cellulosic matrix of the alkaline-extracted SB pretreated by HP was correctly prepared for the enzymatic hydrolysis. Hence, ideal experimental conditions need to correctly combine the levels of the pretreatment variables in the AOP, in order to maximize *EC*. The optimum experimental conditions found in this study, using the Statistica® desirability tool, were applied H_2O_2 load of 0.3 $\text{g}_{\text{H}_2\text{O}_2} \text{g}_{\text{pretreated SB}}^{-1}$, *LSR* of 9.67 mL g^{-1} , and reaction time of 8 h. The use of these conditions resulted in 0.325 g glucose $\text{g}_{\text{SB}}^{-1}$, corresponding to *Y* of 77.96%, *RC* of 82.24%, *EC* of 88.94%, and *OGY* of 60.87%. When the desirable conditions were applied, a relatively low degree of delignification (42.71%) resulted in substantially increased access of the hydrolytic enzymes to the substrate. To demonstrate this, *EC* of 46.8% and *OGY* of 38.5% were achieved after enzymatic hydrolysis of the alkaline-extracted SB pretreated by HP, while the *EC* and *OGY* values increased to 88.94% and 60.87%, respectively, after application of the desirable AOP conditions to alkaline-extracted SB pretreated by HP.

Rabelo et al. (2014) also evaluated the effect of two different *LSR* values on *EC* of the RSF obtained from AOP of raw SB using 3.5 FPU g^{-1} . These authors reported *OGY* of 92.6% for *LSR* of 33.3 mL g^{-1} and *OGY* of 68.6% for *LSR* of 10 mL g^{-1} while *OGY* of 60.9% was obtained in the present study using much lower applied H_2O_2 load.

Zhang et al. (2019) evaluated the *EC* of raw SB pretreated by AOP and reported an *EC* value of 89.1% (*OGY* of 77.6%). However, the applied H_2O_2 load (6.25%) was much higher than that used in the present study (0.3%). In addition, a surfactant (Tween 80) was used in the enzymatic hydrolysis step to improve the *EC* of raw SB pretreated by the AOP. Compared to the results obtained in the present study, the higher applied H_2O_2 load and the addition of Tween 80 in the enzymatic hydrolysis would increase the operating costs on a real scale.

3.4. Morphology of sugarcane bagasse pretreated by advanced oxidative pretreatment

The morphology of the samples of AOP of alkaline-extracted SB pretreated by HP was examined by SEM analysis. SEM micrographs of

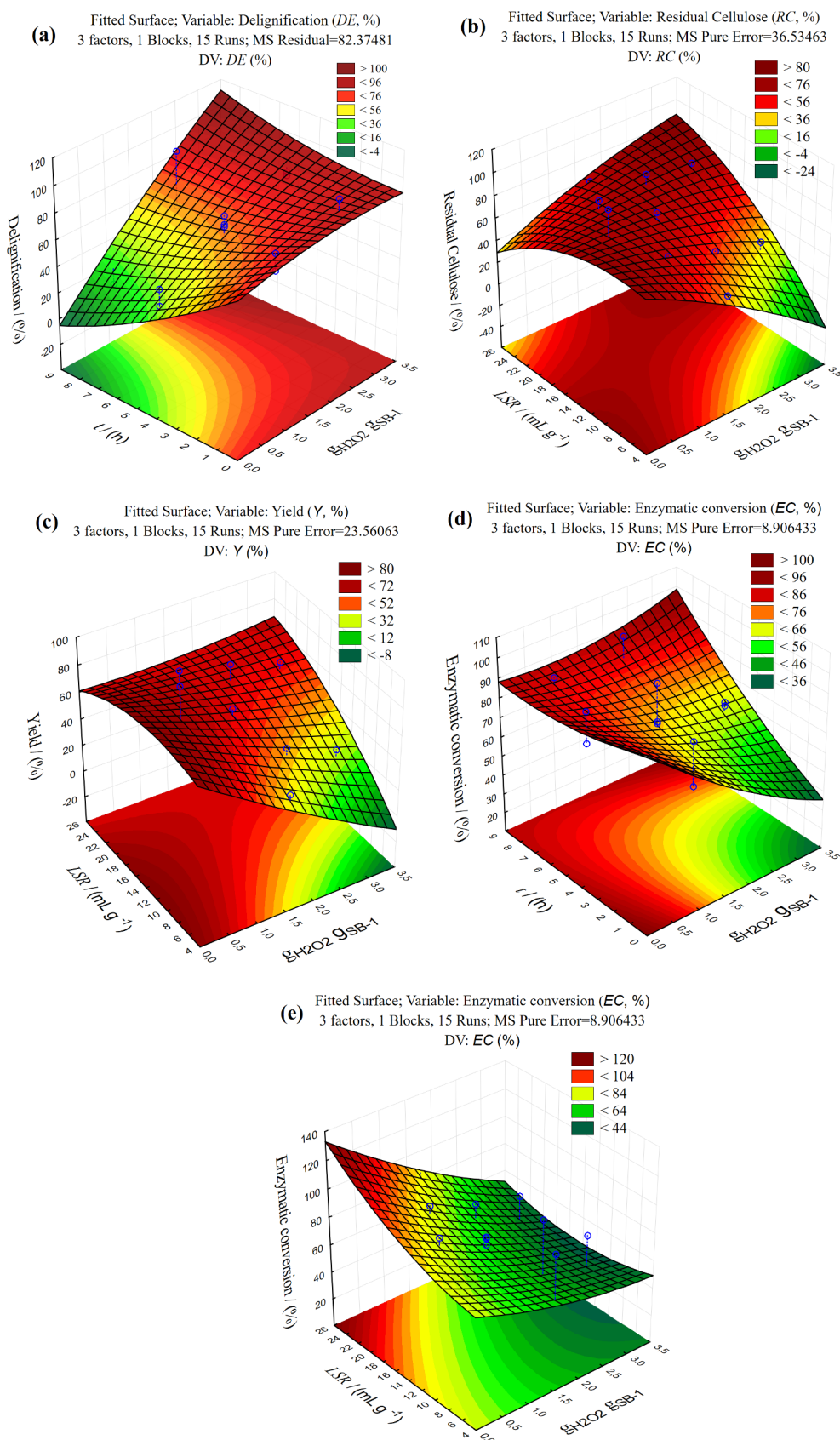


Fig. 1. Response surfaces for (a) delignification of pretreated SB by AOP with LSR fixed at 15 mL g_{SB}⁻¹, (b) residual cellulose (RC) with *t* fixed at 4.5 h, (c) yield (*Y*) with *t* fixed at 4.5 h, (d) enzymatic conversion, as a function of applied H₂O₂ load and time, with LSR fixed at 15 mL g⁻¹, and (e) enzymatic conversion, as a function of LSR and applied H₂O₂ load, with *t* fixed at 1 h.

samples generated from experiments 4, 7, and 12 (Table 2) and desirable AOP conditions are shown in E-supplementary data. The sample generated from experiment 4 presented clean and smooth fiber bundles well disaggregated, as a result of high *DE* value observed. This experimental condition resulted in a high *EC* value because the fiber bundles seem very accessible to hydrolytic enzymes. However, this experimental condition led to low values of *Y* and *RC*, which resulted in a low *OGY* value. The sample generated from experiment 7 presented fiber bundles more aggregated in comparison to those of experiment 4, as a result of the lower *DE* value observed. However, this experimental condition presented *EC* value very similar to that of experiment 4. The same observation was made for that sample generated from experiment 12, which also presented aggregated fiber bundles but the lowest *DE* value observed among all AOP experiments performed. However, this sample also presented a high *EC* value as observed for those samples generated from experiments 4 and 7. The morphology of the sample generated from desirable AOP conditions was very similar to those of experiments 7 and 12. This sample also presented a *EC* value similar to those samples generated from experiments 7 and 12. This suggest that part of the residual lignin in SB pretreated by AOP did not influence on the course of enzymatic hydrolysis, which may be related to the location and physicochemical characteristics of this residual lignin, such as chemical composition and molar mass, as previously discussed. In addition, taking into account the disaggregation of the fibers in the plant cell wall, it seems that the middle lamella lignin has been largely removed, which undoubtedly increased the access of hydrolytic enzymes to the cellulose fibers, which could explain the small difference in *EC* values observed for samples generated by different AOP conditions.

3.5. Second-generation ethanol production

Evaluation was made of 2G ethanol production by fermentation (without addition of a nutrient solution) of the glucose-rich hydrolysate resulting from enzymatic hydrolysis of the solid fraction prepared under the optimum conditions obtained using the Statistica® desirability tool. Three different strains of *Saccharomyces cerevisiae* were used (PE2, CAT1, and LBCM1047). The composition of the hydrolysate was 2.0 g L^{-1} cellobiose, 78.7 g L^{-1} glucose, 5.75 g L^{-1} xylose, and 0.5 g L^{-1} arabinose. No furans (HMF or furfural) were detected in the hydrolysate. In addition to the furans, organic acids may also act as inhibitors of certain yeasts (Araujo et al., 2018; Rabelo et al., 2014). A low concentration of formic acid was detected in the hydrolysate (0.26 g L^{-1}), while no acetic acid was detected.

The high glucose concentration and low xylose and arabinose concentrations, as well as the absence of fermentation inhibitors such as HMF and furfural in the hydrolysate, confirmed that the SB fractionation strategy developed here was successful. Fig. 2a presents the profiles of 2G ethanol production, as a function of fermentation time. The efficiencies of glucose conversion into 2G ethanol were 89.4% for LBCM1047, 81.2% for CAT1, and 82.9% for PE2. These results demonstrated that the LBCM1047 strain presented superior efficiency of conversion of glucose into 2G ethanol, compared to the strains conventionally used in Brazilian sugar mills.

Fig. 3 presents the mass balance for estimation of 2G ethanol production using the SB fractionation strategy proposed in this study, considering the processing of 1.0 kg of raw SB. It was possible to produce $190.8 \text{ L}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$ ($150.5 \text{ kg}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$), corresponding to $466 \text{ L}_{\text{ethanol}} \text{ ton}_{\text{pretreated SB}}^{-1}$ ($367.7 \text{ kg}_{\text{ethanol}} \text{ ton}_{\text{pretreated SB}}^{-1}$), using the RSF obtained by AOP of alkaline-extracted SB pretreated by HP. Martins et al. (2015) reported a conversion of glucose into 2G ethanol of 84.48% ($171.6 \text{ L}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$ or $135.4 \text{ kg}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$), using AOP directly applied to raw SB ($0.51 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, *LSR* of 5 mL g^{-1} , 90°C) and subsequent enzymatic hydrolysis using $15 \text{ FPU g}_{\text{SB}}^{-1}$ and *LSR* of 10 mL g^{-1} . The higher amount of 2G ethanol obtained in the present study was due to the greater access of the enzymes to the substrate in the enzymatic hydrolysis step, which provided a greater amount of

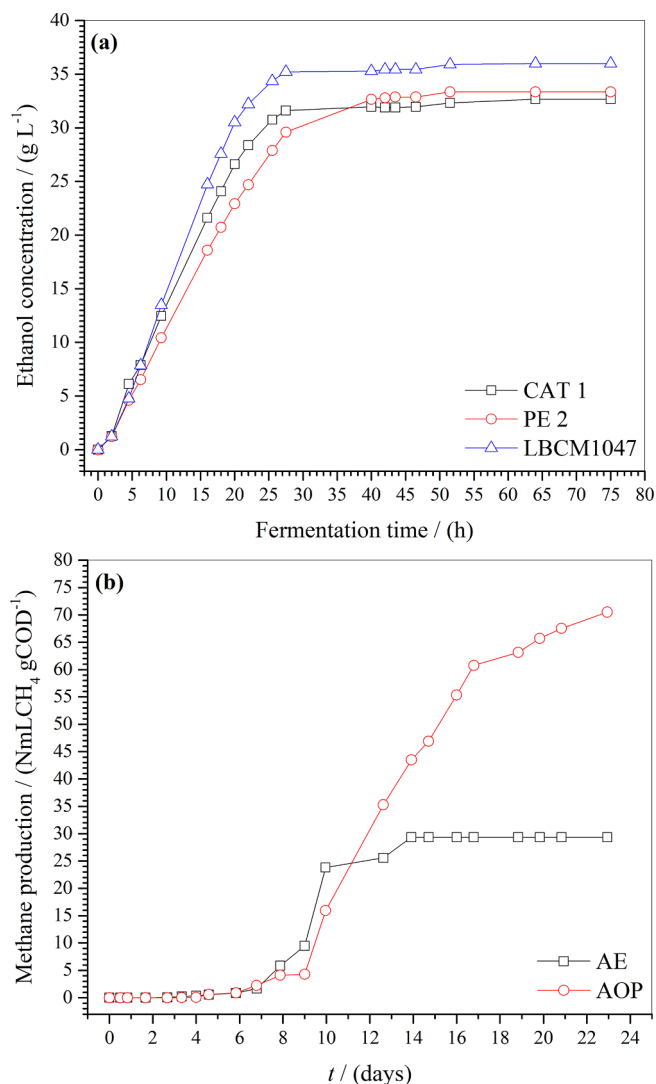


Fig. 2. (a) Fermentation of the hydrolysate obtained from enzymatic hydrolysis of pretreated SB, using different strains of *Saccharomyces cerevisiae*, and (b) methane production from the residual liquid fractions of the alkaline extraction and advanced oxidative pretreatment steps.

fermentable C6 sugars. In addition, the lower severity of the pretreatment steps applied to SB in the present study led to higher *RC*, compared to the value reported by Martins et al. (2015).

Rabelo et al. (2014) evaluated two different solid loadings in enzymatic hydrolysis of the RSF obtained after AOP of raw SB with $2.67 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, producing $176.4 \text{ L}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$ ($139.2 \text{ kg}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$) with *LSR* of 10 mL g^{-1} , and $238.1 \text{ L}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$ ($187.9 \text{ kg}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$) with *LSR* of 33 mL g^{-1} . Therefore, in comparison to the present study, higher 2G ethanol production was only obtained when a much higher *LSR* value was used. However, the use of a high *LSR* in the enzymatic hydrolysis step is economically unfavorable for both the fermentation and the distillation steps in the 2G ethanol production process.

3.6. Biogas production from residual streams of the SB fractionation

After each fractionation step, the residual liquid fractions (RLFs) were submitted to AD for biogas production. The AD of the RLF generated in the HP step of SB processing, using the conditions optimized by Baêta et al. (2016a), yielded $140.0 \text{ NmL}_{\text{CH}_4} \text{ g}_{\text{VSS}}^{-1}$ ($15.16 \text{ NL}_{\text{CH}_4} \text{ kg}_{\text{SB}}^{-1}$ or $0.173 \text{ Nm}^3_{\text{CH}_4} \text{ kg}_{\text{COD}}^{-1}$). The AD of the RLF generated after the AE of SB pretreated by HP yielded $56.3 \text{ NmL}_{\text{CH}_4} \text{ g}_{\text{VSS}}^{-1}$ ($6.3 \text{ NL}_{\text{CH}_4} \text{ kg}_{\text{SB}}^{-1}$) (Fig. 2b). The lower methane production by AD of the RLF generated

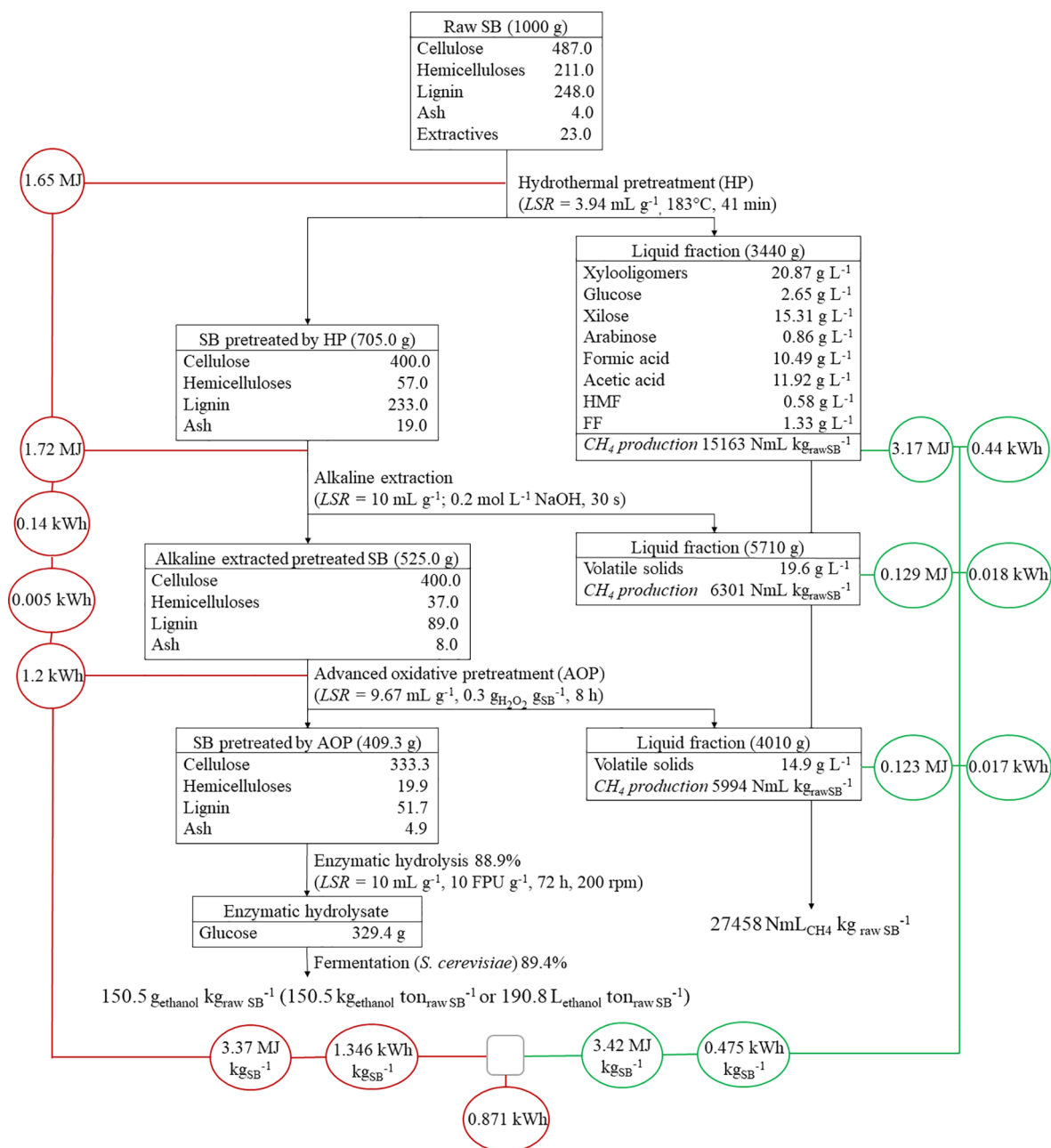


Fig. 3. Mass and energy balances for the SB fractionation strategy proposed in this study.

from the AE step, compared to methane production by AD of the HH, which was reported by Baêta et al. (2016a), was probably due to the high DE achieved in the AE step (61.8%), resulting in a residual liquid stream with a greater amount of lignin fragments of high toxicity. These fragments were likely to hamper and/or decrease the growth of methanogenic microorganisms, as reported by Baêta et al. (2016a), who observed that methane production decreased when a higher concentration of soluble lignin fragments was present in the liquid fraction (hemicelluloses hydrolysate) submitted to AD. For the residual liquid stream from the AOP of alkaline-extracted SB pretreated by HP, the biogas yield was 100.2 NmL_{CH₄} g_{VSS}⁻¹ (5.99 NL_{CH₄} kg_{SB}⁻¹). For both fractionation steps (AE and AOP), the presence of Na⁺ ions (from the NaOH used) in the liquid phase submitted to AD may have caused inhibition or increased the cell wall osmotic pressure of the microorganisms, restricting their growth (Deublein and Steinhauser, 2011).

The total methane production from the AD of the residual liquid streams of the SB fractionation strategy was 296.5 NmL_{CH₄} g_{VSS}⁻¹

(27.46 NL_{CH₄} kg_{SB}⁻¹ or 0.273 Nm³_{CH₄} kg_{COD}⁻¹). Nosratpour et al. (2018) reported values of 239.7 and 130.7 NmL_{CH₄} g_{VSS}⁻¹, using SB pretreated with 0.5 mol L⁻¹ sodium carbonate at 140 and 180 °C, respectively. Ahmad et al. (2018) reported yields of 174.8 and 323.3 NmL_{CH₄} g_{VSS}⁻¹ from AD of a solid fraction resulting from HP of SB impregnated with H₂O₂ at 0.64 and 7.36% (v v⁻¹), respectively.

The biodegradabilities of the RLFs from the AE and AOP fractionation steps were evaluated by comparison of the results obtained in the present work with the maximum theoretical conversion of COD to CH₄ (354.0 NmL_{CH₄} g_{COD}⁻¹ or 0.354 Nm³_{CH₄} kg_{COD}⁻¹). For the AE step, 29.36 NmL_{CH₄} g_{COD}⁻¹ (0.0294 Nm³_{CH₄} kg_{COD}⁻¹) was produced, while the RLF of the AOP produced 70.52 NmL_{CH₄} g_{COD}⁻¹ (0.0705 Nm³_{CH₄} kg_{COD}⁻¹). Fig. 2b presents the methane production from both RLFs (AE and AOP), as a function of time. In the absence of more readily accessible and degradable compounds, a longer time is required for the microorganisms to hydrolyze the compounds solubilized after alkaline extraction and AOP. During this period of time, acidogenic, acetogenic, and especially

methanogenic microorganisms are deprived of nutrients, leading to reduction of their metabolic activities. Compounds with greater biodegradability are more readily hydrolyzed, compared to low digestibility compounds. Therefore, the VFA concentration in the reaction medium tends to remain equilibrated, leading to better functioning of the microbial consortium in the AD system. In addition, the hydrolysis rate is lower in the presence of low biodegradability compounds (Nosratpour et al., 2018).

Although the AOP step increased the methane yield from the RLF, probably due to a decrease in the concentration of phenolic fragments, the RLF samples from both the AE and the AOP steps exhibited low biodegradability (8.4% for AE and 20.15% for AOP). Despite this, the total methane production using the RLFs of all the steps (HP, AE, and AOP) could enhance the energy balance of the SB fractionation strategy proposed in this study.

3.7. Energy balance

The estimated mass and energy balances for the proposed SB fractionation strategy are presented in Fig. 3. The fractionation strategy required 3.37 MJ of thermal energy to process 1.0 kg of SB, consisting of 1.65 MJ for HP of SB and 1.72 MJ for drying the SB pretreated by HP before the AE step. The thermal energy demands of both the HP and the AE steps were met by the thermal energy produced in a CHP system by combustion of the biogas produced from AD ($3.42 \text{ MJ kg}_{\text{SB}}^{-1}$) of all the RLFs. This energy balance showed the importance of the biogas generated from the RLFs of the AE and AOP steps for the fractionation strategy, since the energy balance obtained considering only the thermal energy generated by the combustion of methane produced from AD of the RLF generated in the HP of SB would be negative (-0.2 MJ).

The electric energy required for the proposed fractionation strategy was $1.346 \text{ kWh kg}_{\text{SB}}^{-1}$, which included consumptions of 0.14 kWh by the Wiley mill, 0.005 kWh by the mechanical mixer used to agitate the suspension in the AE step, and 1.2 kWh by the mechanical mixer used to agitate the suspension in the AOP step. The much greater amount of electric energy required for the AOP step, compared to the AE step, was due to the different operation times (30 s for AE and 8 h for AOP). The electric energy generated by the combustion of the biogas produced was equivalent to $0.475 \text{ kWh kg}_{\text{SB}}^{-1}$, resulting in an electric energy deficit of 0.871 kWh for the proposed fractionation strategy. This energy deficit was essentially due to the agitation time (8 h) required for the AOP, under the optimized conditions obtained using the desirability tool. Xue et al. (2016) and Banerjee et al. (2011) applied oxidative pretreatments with H_2O_2 to different lignocellulose biomasses, with reaction times of 24 and 48 h, respectively. However, Rabelo et al. (2014) used a reaction time of 1 h, although a much higher H_2O_2 load ($2.68 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$) was employed, compared to the load used in the present study.

Nonetheless, the electric energy produced was sufficient to ensure operation of the AOP reactor for 2 h. Considering a reaction time of 2 h, high values of Y ($< 90\%$), RC ($> 80\%$) and EC ($> 80\%$) could still be obtained, according to the desirability tool used for prediction. In addition, for scale-up of the fractionation strategy in an integrated 1G/2G ethanol plant, optimization of the designs of the mechanical mixers used in the AE and AOP steps could reduce the electric energy required, since the mechanical mixers used in the present study were not optimized for this purpose. Hence, this may have contributed to higher electric energy consumption in the AE and AOP steps.

4. Conclusions

Optimization of the conditions for advanced oxidative pretreatment (AOP) of sugarcane bagasse ($0.3 \text{ g}_{\text{H}_2\text{O}_2} \text{ g}_{\text{SB}}^{-1}$, $9.67 \text{ mL g}_{\text{SB}}^{-1} \text{ LSR}$, 8 h) pretreated by hydrothermal processing and alkaline extraction (AE) led to considerable savings of H_2O_2 , together with a high degree of cellulose preservation. The AE played an important role in optimization of the process. High values were obtained for enzymatic conversion (89.4%) with overall glucose yield of 60.9%, conversion of glucose to ethanol

(88.9%), and ethanol production ($190.8 \text{ L}_{\text{ethanol}} \text{ ton}_{\text{SB}}^{-1}$). Anaerobic digestion of the liquid streams yielded $27.46 \text{ NL}_{\text{CH}_4} \text{ kg}_{\text{SB}}^{-1}$, providing benefits in terms of both effluent treatment and the energy balance.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2019.121963>.

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